

# Dr. Dipankar Bhattacharya, Ph.D.

**Marie Skłodowska-Curie Postdoctoral Fellow (2025–2027)**

Dyson School of Design Engineering, Imperial College London, United Kingdom

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## Research Profile

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Independent robotics researcher developing model-based and learning-enabled robotic systems for health-care, rehabilitation, and advanced manufacturing. My work integrates soft robotics, cable-driven actuation, deformable-object manipulation, perception, and control, including industry-funded esophageal stent testing on the RoSE platform (BvW Holding, NZD 39,000, 2018). As Marie Skłodowska-Curie Fellow at Imperial College London, I lead CASREx (Grant No. 101205678; EUR 276,187.92, 24 months) on cable-driven exosuits for upper-arm muscle-impairment rehabilitation, building on RoSE (Soft Robotics, 2021), cable wrapping modeling (MMT, 2025; IEEE TCST, 2023), and fabric manipulation (IEEE IROS, 2026; IEEE RA-L, 2025).

## Education

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| <b>2016 – 2021</b> | <b>Ph.D., Mechatronics Engineering</b> in Department of Mechanical and Mechatronics Engineering, The University of Auckland (UoA), 01/08/2016 – 14/04/2021.<br>Thesis title: <i>Robotic Soft Esophagus Design, Modelling, and Control for Endoprosthetic Stent Testing and Food Swallow Investigation</i><br>Developed RoSE for physiologically relevant stent testing and swallow investigation; outputs: 3 journal articles (2 in Soft Robotics, 2 IEEE TIE), 2 conference papers, 1 invited book chapter (Springer, 2019). |
| <b>2011 – 2013</b> | <b>M.Tech , Systems and Control</b> in Department of Electrical Engineering, Indian Institute of Technology (IIT), 22/07/2011 – 04/07/2013.<br>Dissertation title: <i>Non-Linear Predictive Control Model using Relevance Vector Machine</i> . GPA: <b>8.4 / 10.0</b><br>Nonlinear MPC with Relevance Vector Machine, validated on a magnetic-levitation system; 1 journal article (Int. J. Sustainable Energy, 2015).                                                                                                        |
| <b>2004 – 2010</b> | <b>B.Tech , Electronics and Communication Engineering</b> , North Eastern Regional Institute of Science and Technology (NERIST), 21/07/2004 – 23/06/2010.<br>Project title: <i>Design and Synthesis of a Sixth Order Tow Thomas Bi-quad filter</i> . GPA: <b>9.1 / 10.0</b>                                                                                                                                                                                                                                                   |

## Employment History

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| <b>2025 – Present</b> | <b>Marie Skłodowska-Curie Postdoctoral Fellow</b> , Dyson School of Design Engineering, Imperial College London. Supervisor: Prof. Thrishantha Nanayakkara ( <i>Morph Lab</i> ).<br><i>CASREx: Cable-driven Arm Soft Reconfigurable Exoskeleton for Muscle-Impairment Rehabilitation</i> (Grant No. 101205678; EUR 276,187.92, 24 months). <ul style="list-style-type: none"><li>• Leading fellowship project on human-inspired stiffness modulation for cable-driven elbow exosuits for upper arm muscle impairment rehabilitation.</li><li>• Integrating human trials, EMG, digital twins, and reinforcement learning for adaptive exosuit control.</li></ul> |
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- Developing and validating a cable-driven manipulator–exosuit platform with compliant actuation and antagonistic co-contraction control.

- 2026 – Present** **Assistant Supervisor**, Dyson School of Design Engineering, Imperial College London, United Kingdom, 15/01/2026 – Present.  
Supervising two Year 3 MEng (Design Engineering) placement students: **Ben Surtees** and **Lijiangke Liu** (March 2026 – Present). Research and development projects related to exosuit development.  
Delivered lecture on **Robotics Simulation and Control using PyDrake** for **DESE71002 Robotics Research Projects**, Dyson School of Engineering, Imperial College London (28/01/2026).
- 2024–2025** **Senior Research Engineer–Control Systems** in the Center for Transformative Garment Production (TransGP), Mentor: Prof. Kazuhiro Kosuge, 02/07/2024 – 31/10/2025.  
*Physics-based Cloth Manipulation with policy generated from imitation learning.*
- Led multidisciplinary work on dual-arm fabric pre-sewing handling and flattening (mesh perception + imitation learning).
  - Contributed to IEEE IROS (2026) on dual-arm fabric flattening (RTFF), IEEE RA-L (2025) on automated straight-line sewing, and IEEE ICMA (2025) fabric destacking.
  - Built and validated robotic fabric manipulation testbeds for industrial sewing and destacking workflows (May–Oct 2025).
- 2024 – 2025** **Visiting Research Associate** in the Department of Electrical and Electronic Engineering, Hong Kong University (HKU), 02/07/2024 – 31/10/2025.
- 2021 – 2024** **Postdoctoral Fellow** in the Department of Mechanical and Automation, The Chinese University of Hong Kong. Supervisor: Prof. Darwin Lau, 01/03/2021 – 30/06/2024  
*Project title 1: Modeling and control of cable-driven robots with cable wrapping around obstacles and links.*  
*Project title 2: Design of a cable-driven growing vine robot for safer false ceiling inspections.*
- Authored journal papers on cable wrapping (MMT, 2025) and iterative-learning control of hybrid cable-driven robots (IEEE TCST, 2023).
  - Mentored undergraduate final year project on design of a cable-driven vine robot for false-ceiling inspection.
  - Co-authored grant proposals and technical reports; mentored PhD students.
- 2021 – 2022** **Visiting Postdoctoral Fellow** in Laboratoire des Sciences du Numérique de Nantes (LS2N), École Centrale de Nantes (ECN). Supervisors: Prof. Stephane Caro and Prof. Darwin Lau, 26/10/2021 – 05/01/2022.  
*Project title: Iterative-learning control of cable-driven parallel robot.*  
Delivered iterative-learning control experiments on cable-driven parallel robots (Oct 2021 – Jan 2022).
- 2017 – 2020** **Graduate Teaching Assistant**, The University of Auckland, New Zealand, 2017–2020.
- 2018 – 2019** **Research Assistant** at The University of Auckland, under the direction of Prof Peter Xu, 09/01/2018 – 08/01/2019.
- 2018 – 2019** **Research Assistant**, The University of Auckland (Prof. Peter Xu), 09/01/2018 – 08/01/2019.
- Measured migration of esophageal stents of multiple designs on robotics soft esophagus platform (BvW-funded project), and published in Soft Robotics (2021).
  - Developed actuation protocols and test reports for industry stent-testing deliverables.
- 2013 – 2016** **Lecturer (Assistant Professor Grade-1)**, School of Electrical, Electronics and Communication Engineering, Galgotias University, India, 19/07/2013–18/06/2016.

## Teaching Experience

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| 2026        | <b>DESE71002</b> , <i>Robotics Research Projects</i> , Year 4/MSc (Design Engineering), Dyson School of Design Engineering, Imperial College London, United Kingdom.<br>Guest lecture on <i>Robotics Simulation and Control using PyDrake</i> (28/01/2026).                                                                                                                                                                                                                                                                                                                 |
| 2021 – 2024 | <b>MAEG5090</b> , <i>Topics in Robotics</i> , for postgraduate students (MSc/MPhil), Department of Mechanical and Automation Engineering, The Chinese University of Hong Kong, Hong Kong SAR. Class size: 50 students.                                                                                                                                                                                                                                                                                                                                                      |
| 2017 – 2020 | <b>MECHENG 706</b> , <i>Mechatronics Design Projects</i> , for 4th year undergraduate students, The University of Auckland, New Zealand.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 2017 – 2020 | <b>MECHENG 306</b> , <i>Design of Sensing and Actuating Systems</i> , for 3rd year undergraduate students, The University of Auckland, New Zealand.                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 2017 – 2020 | <b>ENGGEN 115</b> , <i>Principles of Engineering Design</i> , for undergraduate students, The University of Auckland, New Zealand.                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 2017 – 2020 | <b>ENGGEN 131</b> , <i>Introduction to Engineering Computation and Software Development</i> , for undergraduate students, The University of Auckland, New Zealand.                                                                                                                                                                                                                                                                                                                                                                                                          |
| 2013 – 2016 | Undergraduate courses (class size: 60 each), School of Electrical, Electronics and Communication Engineering, Galgotias University, India:<br><b>BECE3304</b> <i>Neural Networks and Fuzzy Control</i> ; <b>BECE2010</b> <i>Digital Electronics</i> ; <b>BECE2016</b> <i>Signals and Systems</i> ; <b>BTEE2002</b> <i>Network Analysis and Synthesis</i> ; <b>BEEE3002</b> <i>Control Systems</i> ; <b>BEEE2001</b> <i>Electrical Measurement and Instrumentation</i> ; <b>BECE3020</b> <i>Digital Signal Processing</i> ; <b>BTEE3019</b> <i>Advanced Control System</i> . |
| 2013 – 2016 | Postgraduate courses (class size: 20), School of Electrical, Electronics and Communication Engineering, Galgotias University, India:<br><b>BEEE5004</b> <i>Digital Control</i> .                                                                                                                                                                                                                                                                                                                                                                                            |

## Research Publications

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### Journal (Published/Accepted)

- [1] B. Jin\*, A. Kobayashi\*, D. Bhattacharya, A. Seino, F. Tokuda, N. Tien, and K. Kosuge. “Automated Straight-line Sewing of Stretchable Fabrics with Different Lengths”. In: *IEEE Robotics and Automation Letters* (2025). \*Equal contribution., pp. 1–8. DOI: [10.1109/LRA.2025.3619791](https://doi.org/10.1109/LRA.2025.3619791).
- [2] D. Bhattacharya, T. K. Cheung, Y. Wang, and D. Lau. “Kinematic and Dynamic Modeling of Cable-Object Interference and Wrapping in Complex Geometrical-Shaped Cable-Driven Parallel Robots”. In: *Mechanism and Machine Theory* 214 (2025), p. 106092. DOI: [10.1016/j.mechmachtheory.2025.106092](https://doi.org/10.1016/j.mechmachtheory.2025.106092).
- [3] D. Bhattacharya, Y. P. Chan, S. Shang, Y. S. Chan, Y. Tan, and D. Lau. “Tri-Space Operational Control of Redundant Multilink and Hybrid Cable-Driven Parallel Robots Using an Iterative-Learning based Reactive Approach”. In: *IEEE Transactions on Control Systems and Technology* (2023). DOI: [10.1109/TCST.2023.3263689](https://doi.org/10.1109/TCST.2023.3263689).
- [4] D. Bhattacharya, L. K. Cheng, and W. Xu. “Nonlinear Model Predictive Control of a Robotic Soft Esophagus for Endoprosthetic Stent Testing.” In: *IEEE Transactions on Industrial Electronics* (2021). DOI: [10.1109/TIE.2021.3121755](https://doi.org/10.1109/TIE.2021.3121755).
- [5] D. Bhattacharya, S. Jesna, L. K. Cheng, and W. Xu. “RoSE: A Robotic Soft Esophagus for Endoprosthetic Stent Testing.” In: *Soft Robotics* (2021). DOI: [10.1089/soro.2019.0205](https://doi.org/10.1089/soro.2019.0205).

- [6] Y. Dang, Y. Liu, R. Hashem, D. Bhattacharya, J. Allen, M. Stommel, L. K. Cheng, and W. Xu. “SoGut: A Soft Robotic Gastric Simulator”. In: *Soft Robotics* (2021). DOI: [10.1089/soro.2019.0136](https://doi.org/10.1089/soro.2019.0136).
- [7] D. Bhattacharya, L. K. Cheng, and W. Xu. “Sparse Machine Learning Discovery of Dynamic Differential Equation of an Esophageal Swallowing Robot”. In: *IEEE Transactions on Industrial Electronics* 67.6 (2020), pp. 4711–4720. DOI: [10.1109/TIE.2019.2928239](https://doi.org/10.1109/TIE.2019.2928239).
- [8] D. Bhattacharya, M. G. Nisha, and G. Pillai. “Relevance vector-machine-based solar cell model”. In: *International Journal of Sustainable Energy* 34.10 (2015), pp. 685–692. DOI: [10.1080/14786451.2014.885030](https://doi.org/10.1080/14786451.2014.885030).

## Books and Chapters

- [9] D. Bhattacharya, L. K. Cheng, S. Dirven, and W. Xu. *Artificial Intelligence Approach to the Trajectory Generation and Dynamics of a Soft Robotic Swallowing Simulator*. Cham: Springer International Publishing, 2019, pp. 3–16. DOI: [10.1007/978-3-319-78452-6\\_1](https://doi.org/10.1007/978-3-319-78452-6_1).

## Conference Proceedings

- [10] D. Bhattacharya\*, K. Tang\*, X. Hang, F. Tokuda, N. Tien, and K. Kosuge. “RTFF: Random-to-Target Fabric Flattening Policy Using Dual-Arm Manipulator”. In: *2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. accepted. \*Equal contribution. 2026.
- [11] W. Dong, D. Bhattacharya, A. Kobayashi, A. Seino, F. Tokuda, X. Huang, K. Tang, N. C. Tien, and K. Kosuge. “Precise Top-layer Fabric Segmentation for Fabric Destacking with Edge- and Shape-Aware Deep Networks”. In: *2025 IEEE International Conference on Mechatronics and Automation (ICMA)*. 2025, pp. 1343–1348.
- [12] D. Bhattacharya, L. K. Cheng, and W. Xu. “Testing and Analysis of Migration Displacement of a Flared Stent Deployed in an Esophageal Swallowing Robot.” In: *25th International Conference on Mechatronics and Machine Vision in Practice (M2VIP)*. 2018, pp. 1–6. DOI: [10.1109/M2VIP.2018.8600911](https://doi.org/10.1109/M2VIP.2018.8600911).
- [13] D. Bhattacharya, L. K. Cheng, S. Dirven, and W. Xu. “Actuation planning and modeling of a soft swallowing robot”. In: *24th International Conference on Mechatronics and Machine Vision in Practice (M2VIP)*. 2017, pp. 1–6. DOI: [10.1109/M2VIP.2017.8211476](https://doi.org/10.1109/M2VIP.2017.8211476).

## In Review/Preparation

- [14] W. Wang, D. Bhattacharya, A. Kobayashi, A. Seino, F. Tokuda, N. Tien, and K. Kosuge. *Automated Fabric Alignment Using Global-Local Optimization and Point Cloud Registration, IEEE Transactions on Automation Science and Engineering (T-ASE)*. in review.
- [15] Y. Wang, E. Chow, D. Bhattacharya, and D. Lau. *Dual-Stage Spatial and Continuum Cable-Driven Parallel Robot for Airlifting in Bored Pile Construction Automation in Construction*. in preparation. 2026.

## Grants and Scholarships

### Grants/Fellowships

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| <b>2025</b> | Recipient, <b>Horizon Europe Marie Skłodowska-Curie Postdoctoral Fellowship</b> for the project “CASREx” (Grant No. 101205678), hosted at <b>Imperial College London</b> . Funded under HORIZON-MSCA-2024-PF-01-01. Duration: 24 months; value: <b>EUR 276,187.92</b> . |
| <b>2023</b> | Co-author, Innovation and Technology Fund (ITF) Partnership Research Programme (PRP) grant for the project “Flexible Robotic Palm for Human Comforting Touching Sense.”, hosted at the Chinese University of Hong Kong (CUHK).                                          |

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| 2022                | Co-authored Research Talent Hub for Innovation and Technology Fund (RTH-ITF) grant for the project “Hybrid Cable-Driven with Active Soft Backbone Continuum Robot for Dexterous Manipulation in Service Applications.” |
| 2018                | Recipient, Dynamic Testing of Implant Stents Using Robotic Soft Esophagus (RoSE), BvW Holding, Cham. Duration: 12 months; value: <b>NZD 39,000</b> .                                                                   |
| 2018, 2019          | Recipient, Riddet Travel Grant, Riddet Institute Centre of Research Excellence (NZD 3,000).                                                                                                                            |
| <b>Scholarships</b> |                                                                                                                                                                                                                        |
| 2019–2020           | Recipient, Doctor of Philosophy Scholarship, University of Auckland.                                                                                                                                                   |
| 2016–2019           | Recipient, Doctor of Philosophy Scholarship, Riddet Institute Centre of Research Excellence.                                                                                                                           |
| 2011–2013           | Recipient, Master of Technology Scholarship, Ministry of Human Resource Development (MHRD), Government of India.                                                                                                       |

## Academic Awards, and Miscellaneous Achievements

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### Academic Awards

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| 15/12/25 | Selected, <b>European Talent Academy 2026</b> (Imperial College London, TU Munich, Politecnico di Milano).                                               |
| 2017     | Recipient, <b>Best Conference paper award</b> , 5th International Conference on Robotic Intelligence Technology and Applications (RiTA), Daejeon, Korea. |
| 2017     | <b>Ranked 2nd</b> in poster presentation in Delivery of Functionality in Complex Food Systems (DoF 2017) Conference.                                     |

### Competitive Examinations

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| 2011 | <b>Percentile: 99.12% (Rank 1,211/1,37,853)</b> in Graduate Aptitude Test of Engineering (GATE). |
| 2010 | <b>Percentile: 96.00%</b> in Graduate Aptitude Test of Engineering (GATE).                       |
| 2004 | <b>Ranked 31st</b> in All India Junior mathematics Olympiad.                                     |

### Professional Activities

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| 05/02/26       | Invited Speaker, <b>Dyson Engineering Departmental Seminar</b> , Imperial College London.                                                                                          |
| 29/01/26       | Invited Speaker, <b>Imperial Robotics Postgrad Committee</b> , Imperial College London. Talk: <i>Learning-based Fabric Manipulation with Mesh Perception and Visual Servoing</i> . |
| 28/01/26       | Delivered lecture on <b>Robotics Simulation and Control using PyDrake</b> for DESE71002 Robotics Research Projects, Dyson School of Engineering, Imperial College London.          |
| 08/08/25       | Invited Speaker, <b>World Robotics Conference (WRC 2025)</b> , Beijing, China. Talk: <i>Fabric modeling and perception for garment manufacturing</i> .                             |
| 03/08/25       | Workshop Organizer and Presenter, <b>IEEE ICMA 2025</b> , Beijing, China. Talk: <i>Mesh-based Fabric State Estimation and Manipulation</i> .                                       |
| 29/05/23       | Poster Presentation, <b>IEEE ICRA 2023</b> , ExCeL London, United Kingdom.                                                                                                         |
| 2018           | Presenter, <b>BioMouth 2018 Meet</b> , Auckland, New Zealand.                                                                                                                      |
| 2021 - Current | Reviewer of IEEE Robotics and Automation Letters (RAL), IEEE International Conference                                                                                              |

on Robotics and Automation (ICRA), ASME Journal of Mechanisms and Robotics (JMR).

- 2017 - Current** Member of the Institute of Electrical and Electronics Engineers (IEEE).
- 2014** Former member of the Electrical Engineering Society (EES), IIT Roorkee, and NERIST Electronics Society (NES), NERIST.
- 2006** Have an 'Outstanding Certificate' in National Service Scheme (NSS) for contributing in activities like blood donation camps, village cleanliness programmes and children education.

## Skills

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- Languages** English and Hindi: professional reading, writing, and speaking proficiency.
- Research areas** Soft robotics, cable-driven robots, robotic perception, deformable-object manipulation, digital twin and control, learning-based modeling and control, and robotic systems for healthcare and manufacturing.
- Programming** Python, MATLAB, PyTorch, ROS2, LabVIEW, assembly-level programming (8085, 8086, MASM32, and 8051), Git, and LaTeX.
- Machine learning** Imitation learning, reinforcement learning, sparse identification modelling, nonlinear MPC with learned dynamics, mesh-based deep perception for deformable objects.
- Simulation** NVIDIA Isaac Sim, pydrake, LeRobot, OpenSim
- CAD and others** Onshape, Blender, SOLIDWORKS, Autodesk Inventor, Inkscape, and Lightroom.
- Hardware** NVIDIA Jetson, Raspberry Pi, myRIO, and Arduino.